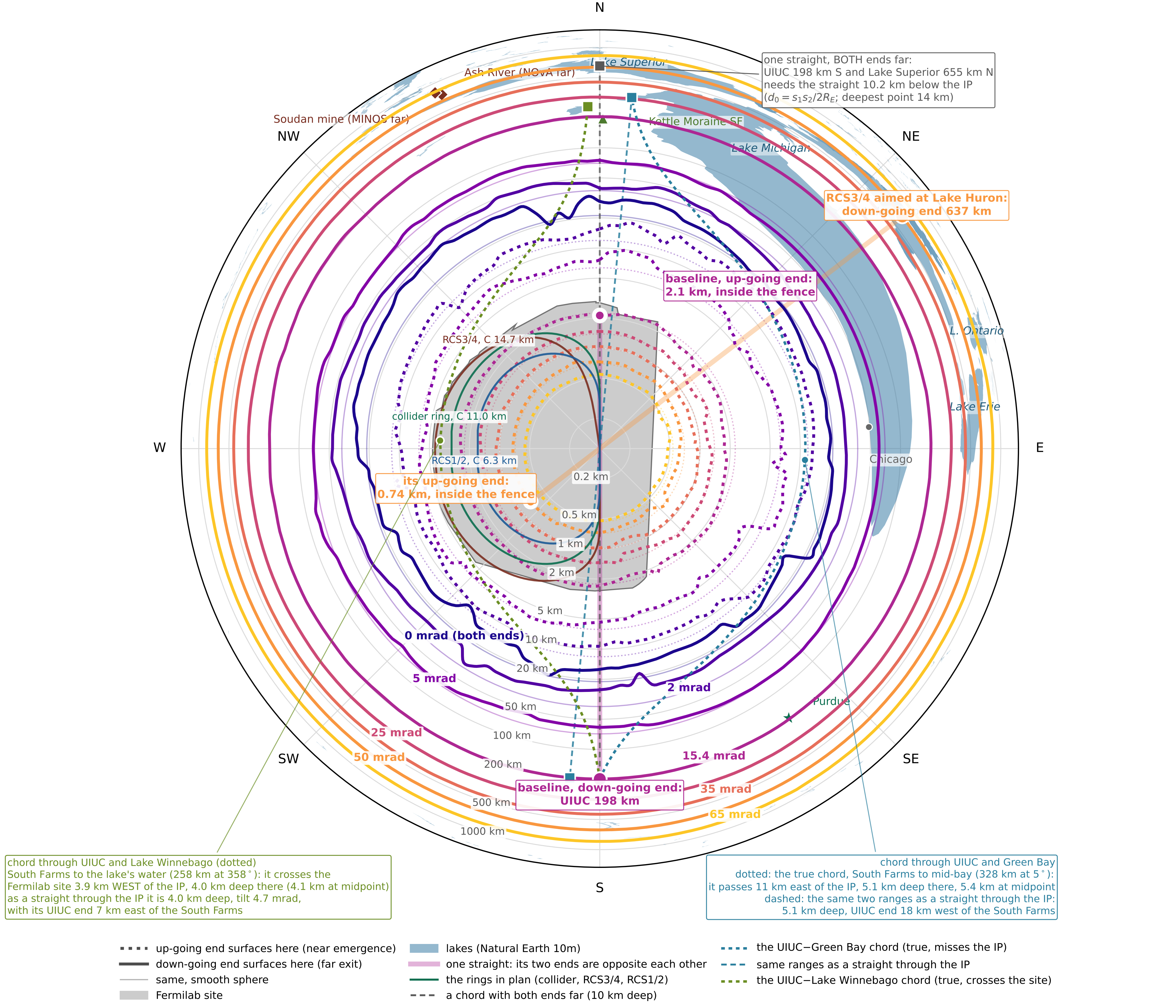
Where the two ends of one straight surface, tilt by tilt

35 m-deep straight; one colour per tilt − dotted: the end that goes up, solid: the end that goes down; at 0 mrad they are the same curve



Bold curves: USGS NED 10 m terrain inside 6 km and GEBCO 2020 beyond, along 144 bearings (0.1−1000 km), Catmull-Rom smoothed; exits over the Great Lakes taken at the water surface. Faint circles: smooth sphere, $s_{near} = R_E(-\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx d_0/\theta$ and $s_{far} = R_E(\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx 2R_E\theta$. Log range; rings in km. The fence is 0.9−4.9 km out, so the dotted curves show which tilts emerge on site.